

MAE 1001



Introduction to Mechanical and Aerospace Engineering

Course website: <https://gwu-mae1001.github.io/fall2025/>

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School of Engineering
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THE GEORGE WASHINGTON UNIVERSITY

Fall 2025

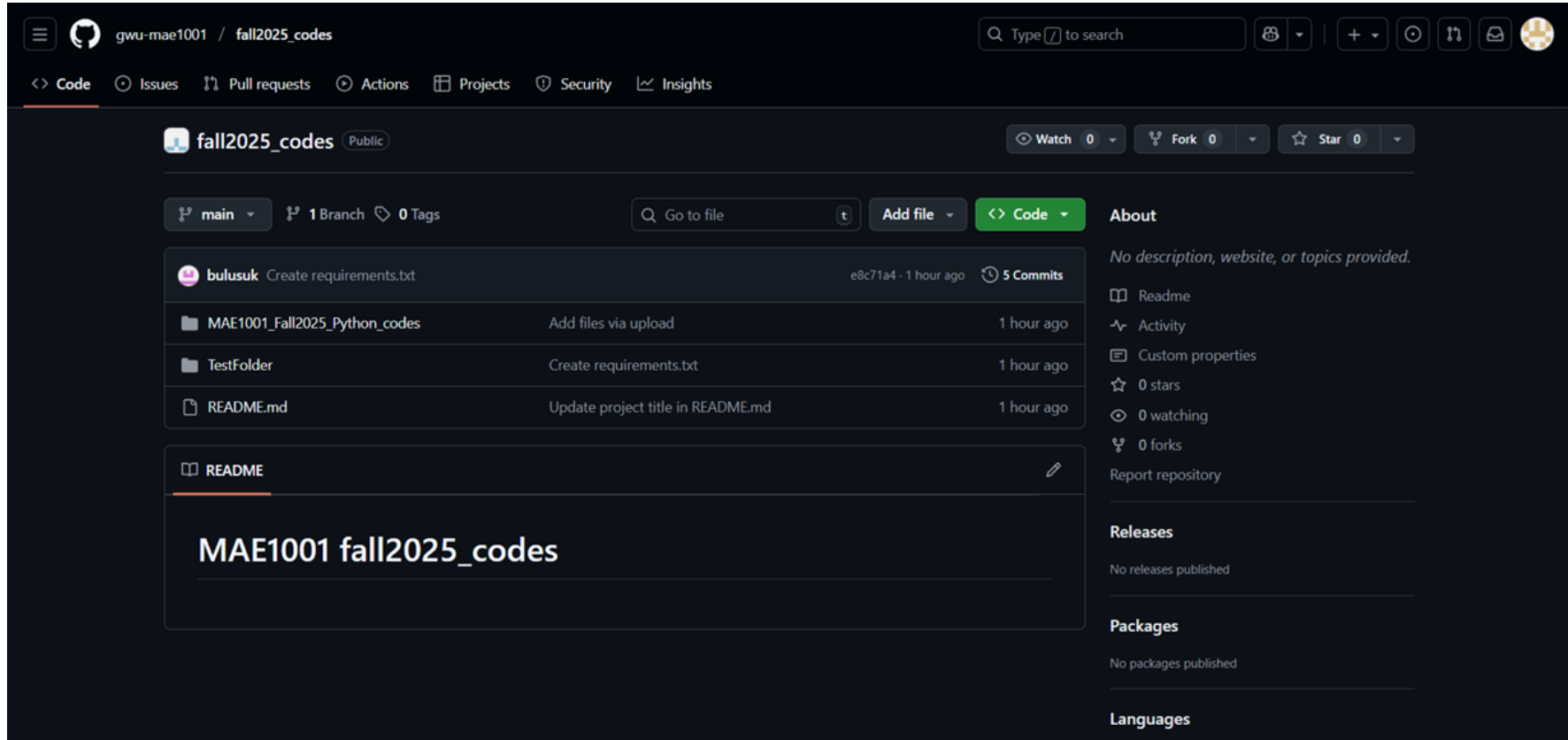
Photo: Kartik Bulusu

Connecting to GitHub Repo and Opening Jupyter Notebook through Anaconda



Step 1

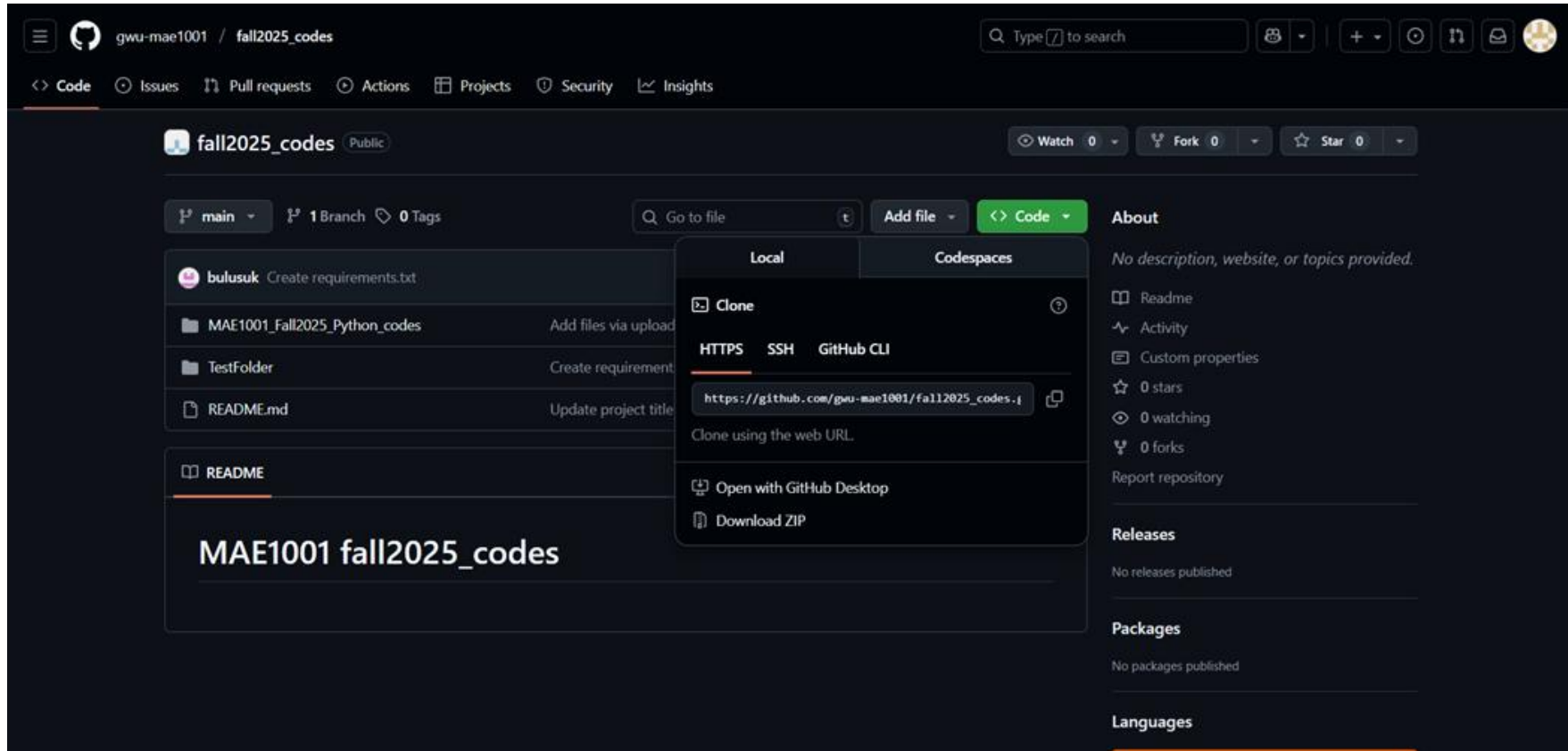
https://github.com/gwu-mae1001/fall2025_codes.git



Access the Class GitHub Repo to access the python codes you will need for the MAE1001 Fall 2025 Semester.



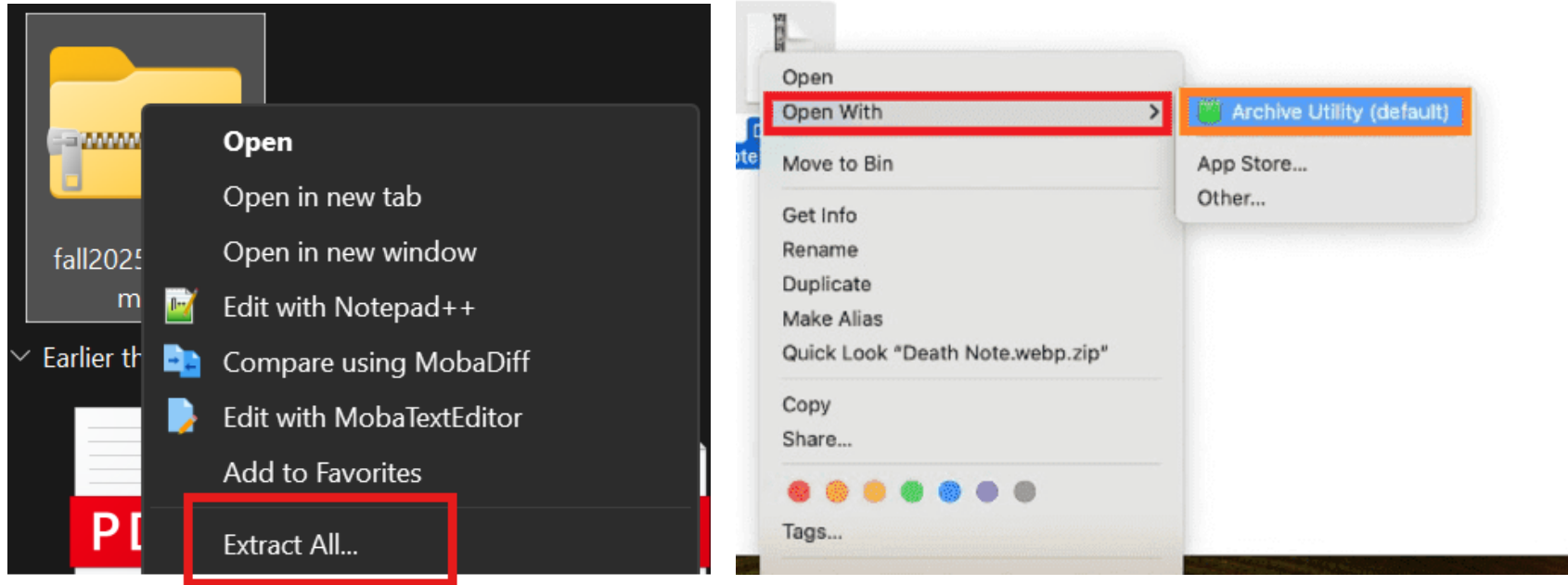
Step 2



Click on 'Code' and select the 'Download ZIP' option to download the python file.



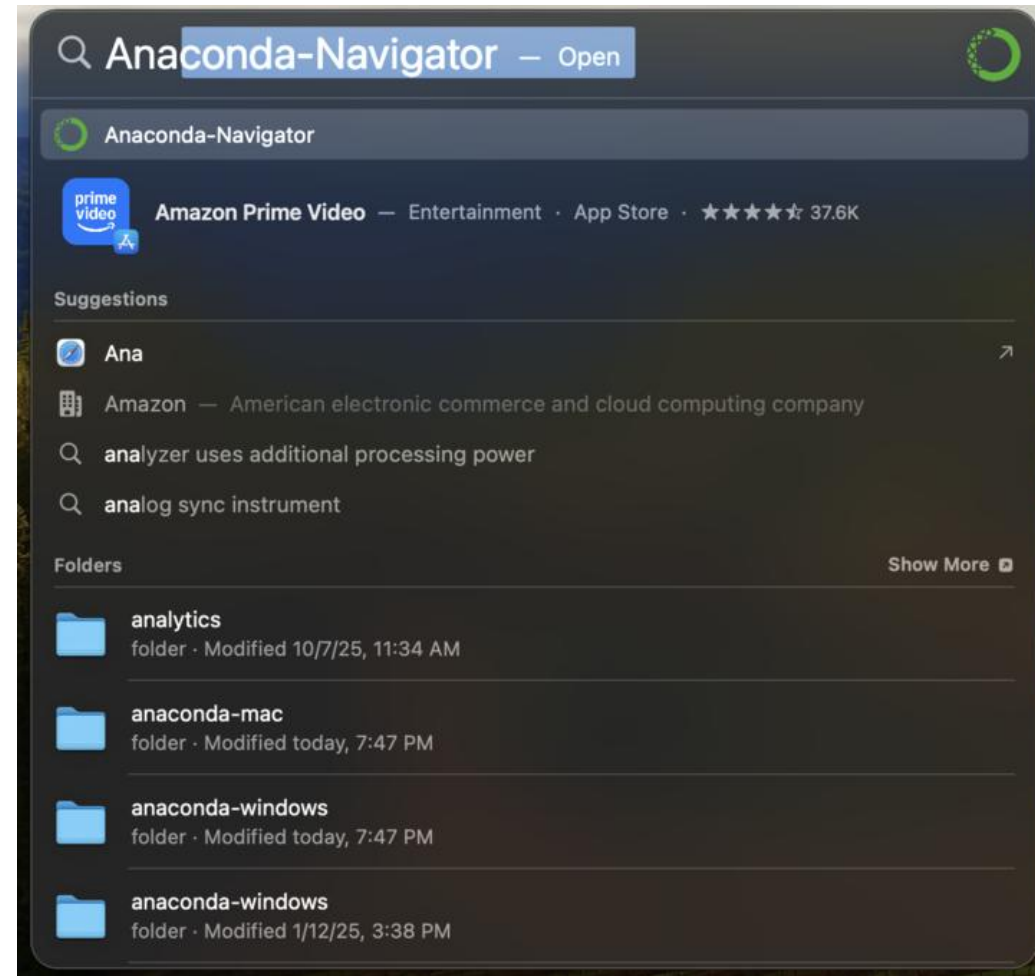
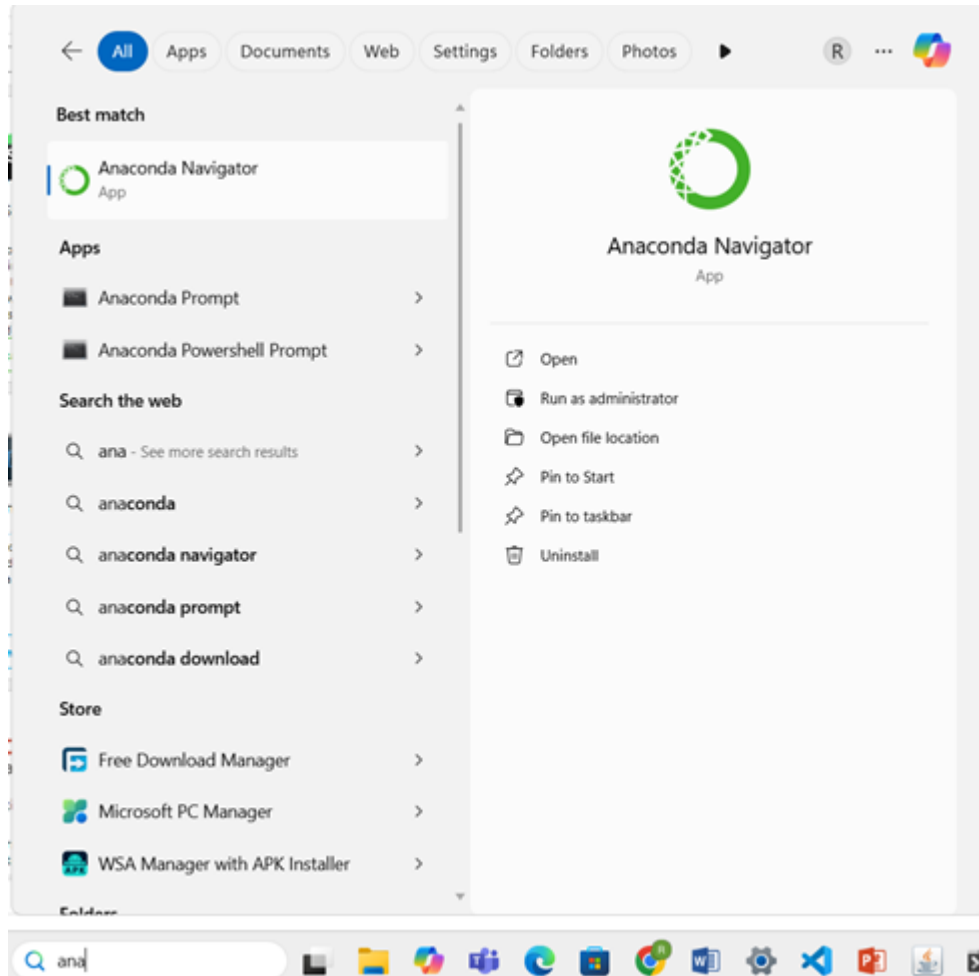
Step 3



Once saved in your Downloads folder, unzip it.



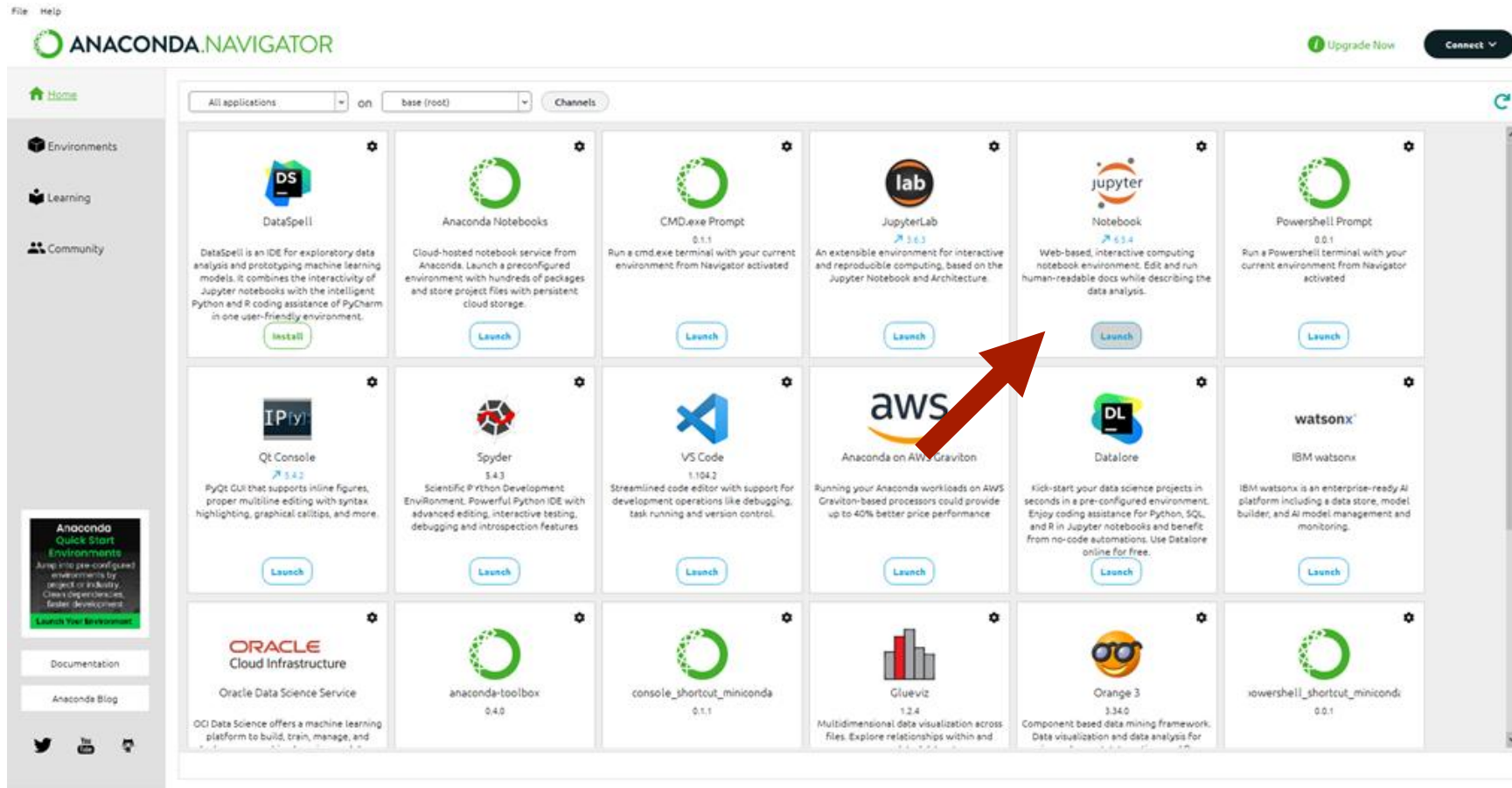
Step 4



Search for Anaconda Navigator in the search bar (Windows) or Spotlight Search (Macbook).



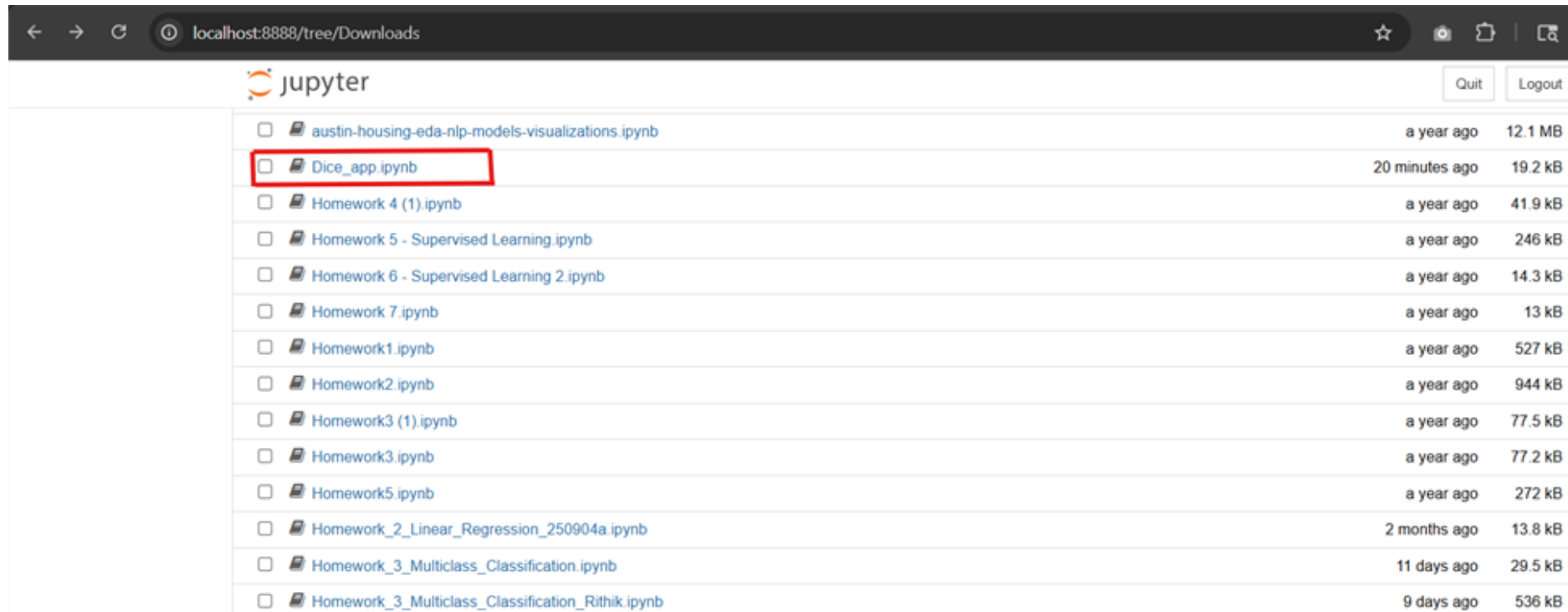
Step 5



Launch Jupyter Notebook to run the python script



Step 6



The screenshot shows a web browser window at localhost:8888/tree/Downloads. The JupyterLab interface displays a list of files in the Downloads directory. The file 'Dice_app.ipynb' is highlighted with a red box. The table below represents the data shown in the file list.

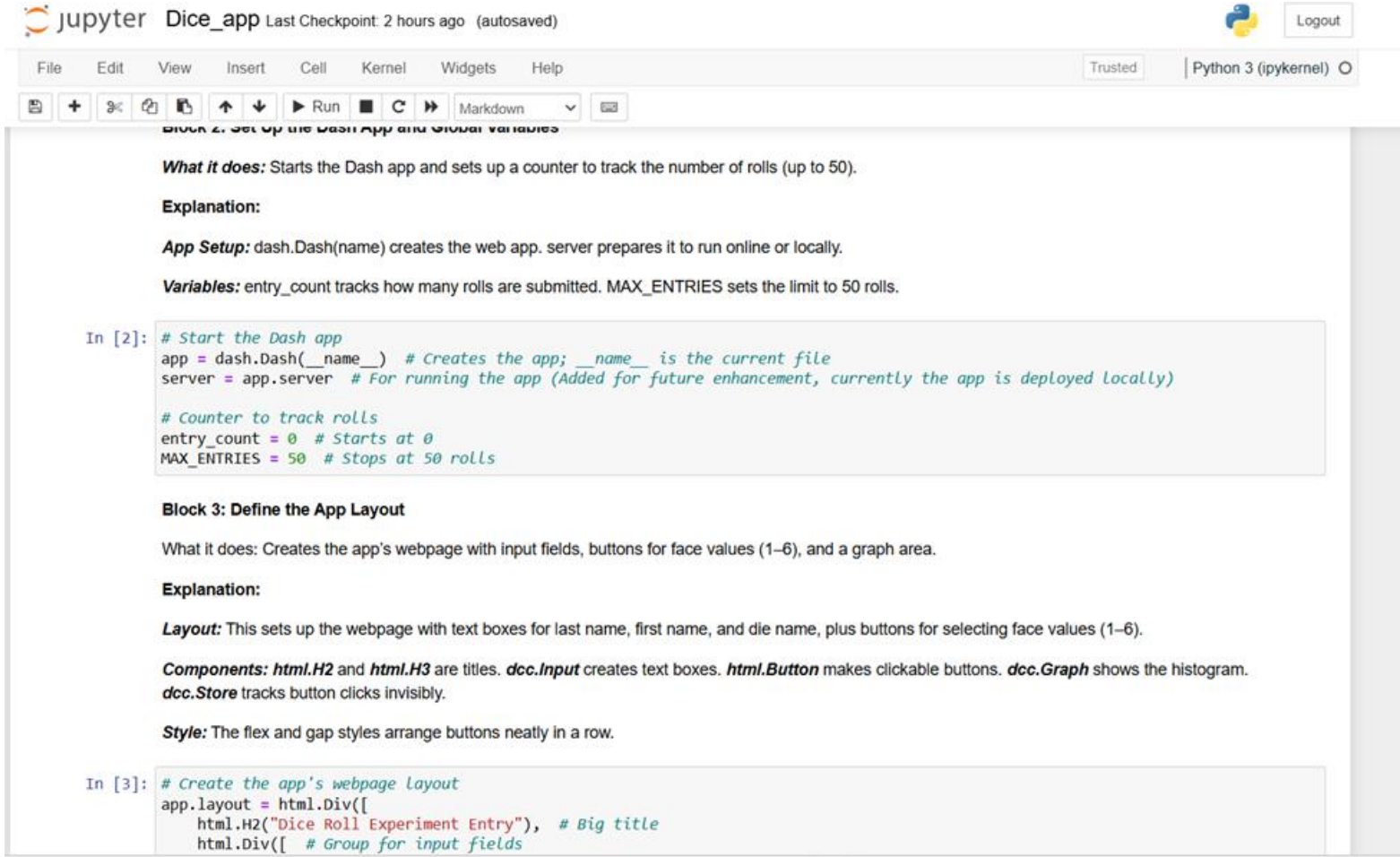
File Name	Time	Size
☐ austin-housing-eda-nlp-models-visualizations.ipynb	a year ago	12.1 MB
☐ Dice_app.ipynb	20 minutes ago	19.2 kB
☐ Homework 4 (1).ipynb	a year ago	41.9 kB
☐ Homework 5 - Supervised Learning.ipynb	a year ago	246 kB
☐ Homework 6 - Supervised Learning 2.ipynb	a year ago	14.3 kB
☐ Homework 7.ipynb	a year ago	13 kB
☐ Homework1.ipynb	a year ago	527 kB
☐ Homework2.ipynb	a year ago	944 kB
☐ Homework3 (1).ipynb	a year ago	77.5 kB
☐ Homework3.ipynb	a year ago	77.2 kB
☐ Homework5.ipynb	a year ago	272 kB
☐ Homework_2_Linear_Regression_250904a.ipynb	2 months ago	13.8 kB
☐ Homework_3_Multiclass_Classification.ipynb	11 days ago	29.5 kB
☐ Homework_3_Multiclass_Classification_Rithik.ipynb	9 days ago	536 kB

Once you have launched Jupyter notebook, navigate to the python file that you have downloaded from github and open it.

The file can be found in the location where you have downloaded the zip file.



Step 7



The screenshot shows a Jupyter Notebook titled "Dice_app" with a last checkpoint 2 hours ago. The interface includes a menu bar (File, Edit, View, Insert, Cell, Kernel, Widgets, Help) and a toolbar with icons for file operations, running cells, and markdown editing. The notebook content is divided into three blocks:

- Block 2: Set up the Dash App and Global variables.**
 - What it does:** Starts the Dash app and sets up a counter to track the number of rolls (up to 50).
 - Explanation:**
 - App Setup:** `dash.Dash(name)` creates the web app. server prepares it to run online or locally.
 - Variables:** `entry_count` tracks how many rolls are submitted. `MAX_ENTRIES` sets the limit to 50 rolls.
- Block 3: Define the App Layout**
 - What it does:** Creates the app's webpage with input fields, buttons for face values (1–6), and a graph area.
 - Explanation:**
 - Layout:** This sets up the webpage with text boxes for last name, first name, and die name, plus buttons for selecting face values (1–6).
 - Components:** `html.H2` and `html.H3` are titles. `dcc.Input` creates text boxes. `html.Button` makes clickable buttons. `dcc.Graph` shows the histogram. `dcc.Store` tracks button clicks invisibly.
 - Style:** The flex and gap styles arrange buttons neatly in a row.

The code cells are as follows:

```
In [2]: # Start the Dash app
app = dash.Dash(__name__) # Creates the app; __name__ is the current file
server = app.server # For running the app (Added for future enhancement, currently the app is deployed locally)

# Counter to track rolls
entry_count = 0 # Starts at 0
MAX_ENTRIES = 50 # Stops at 50 rolls

In [3]: # Create the app's webpage layout
app.layout = html.Div([
    html.H2("Dice Roll Experiment Entry"), # Big title
    html.Div([ # Group for input fields
```

Once you have opened the python file, you can run each code block using Shift + Enter.

On running the program, it saves a CSV file with your results in the same folder as the program.

